Proof Strategy: Partial Compression and the Sign-Reversing Involution

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1 Setup and notation

Let $c = (c_0, c_1, c_2)$ be a rank-3 profile with $d = c_0 + c_1 + c_2$ and $3 \nmid d$. The Q-polynomial is

$$Q_n = \sum_{S \subseteq \{1,\ldots,n\}} (-1)^{|S|}\, q^{\sigma(S)}\, h_{n-|S|},$$

where $h_m = (q)_m \cdot g_m \in \mathbb{N}[q]$ and $\sigma(S) = \sum_{s \in S} s$. Expanding $(q)_m = \sum_{U \subseteq \{1,\ldots,m\}} (-1)^{|U|} q^{\sigma(U)}$, the full ground set is

$$Q_n = \sum_{\substack{S \subseteq \{1,\ldots,n\} \\ U \subseteq \{1,\ldots,m\} \\ \mu \in \mathrm{CPP}_m(c)}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\mu|},$$

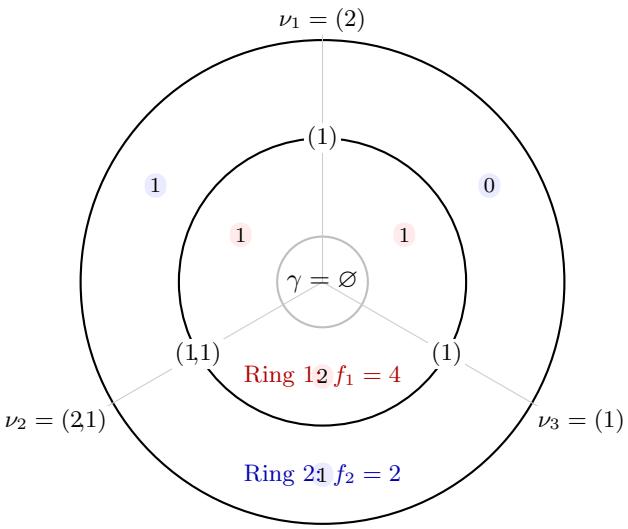
where $m = n - |S|$ and $\mathrm{CPP}_m(c)$ denotes CPPs with profile c and $\max(\mu) = m$.

2 Two views of a cylindric plane partition

A CPP ν with max entry n and profile $c = (c_0, c_1, c_2)$ lives on a cylindric growth diagram. The growth diagram encodes the same data in two equivalent ways: as concentric rings (the *onion*), or as a tower of horizontal bands (the *NILP tower*). We describe both views and the dictionary between them, using the running example $c = (2, 1, 0)$, $\lambda = \big((2), (2,1), (1)\big)$ with max $= 2$ and $|\lambda| = 6$.

2.1 View 1: The onion (concentric rings)

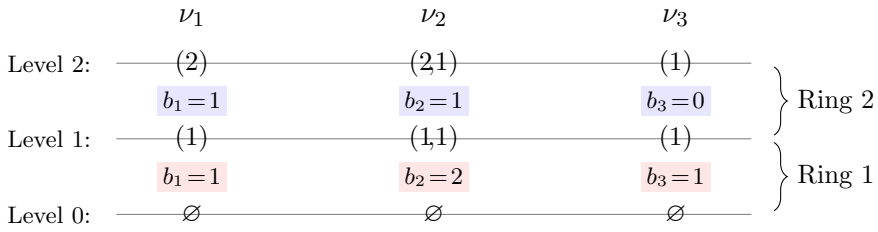
The growth diagram has n concentric *rings*, numbered 1 (innermost) to n (outermost). Each ring carries d face labels—one per face around the cylinder. The boundary between consecutive rings is a triple of partitions (the vertex labels of the growth diagram at that level).



Reading from outside in: the outermost boundary carries the CPP λ; the innermost boundary is the base partition γ. Each ring's face labels encode one "layer" of the CPP—how the boundary partitions change from one level to the next via the Burge local rule. **Peeling** ring k means applying the inverse Burge rule at all d faces of ring k, extracting the face labels and collapsing the boundary inward.

2.2 View 2: The NILP tower (horizontal bands)

The same data can be "unrolled" into a tower of horizontal bands. Band k occupies the strip between levels $k-1$ and k. The d face labels sit inside each band, and the boundary partition triples label the horizontal lines.



The columns correspond to the three partitions ν_1, ν_2, ν_3. Reading upward from the base, each band adds one ring's worth of structure. The partition boundaries at each level trace *paths* running vertically through the tower: these are the non-intersecting lattice paths on the cylinder, and the interlacing conditions ensure they never cross.

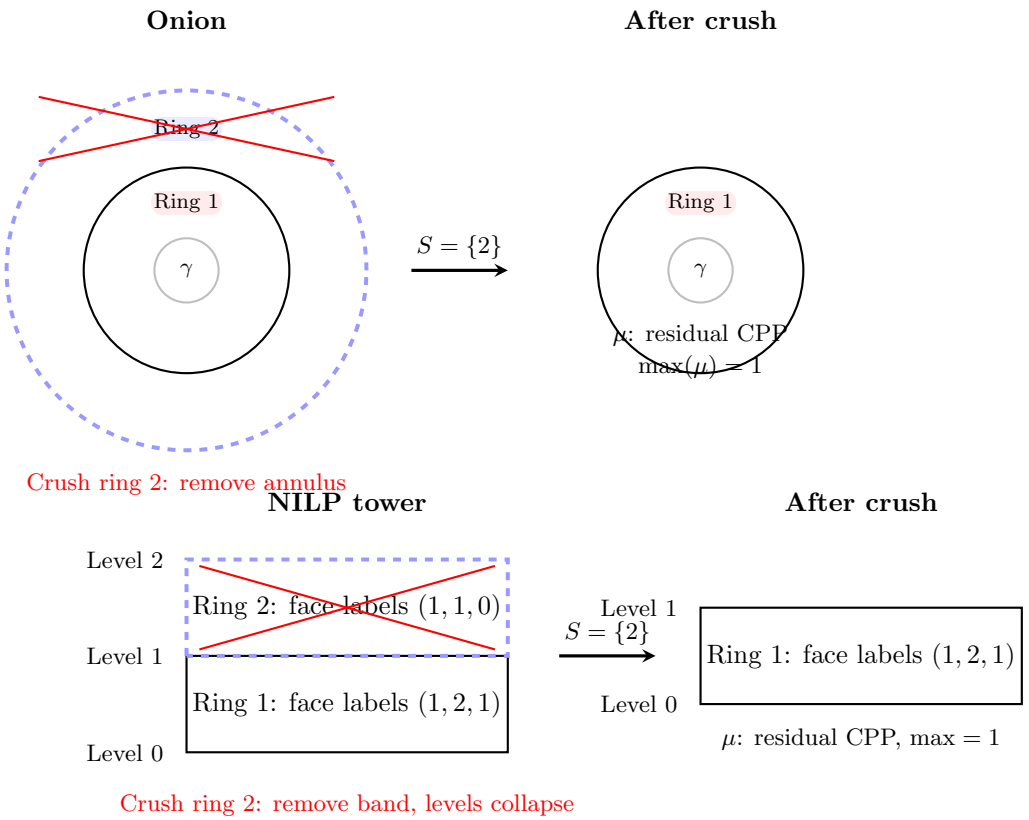
2.3 The dictionary

Concentric rings (onion)	NILP tower (bands)
Ring k (annular region)	Horizontal band at height k
Face labels at ring k	Cell contents in band k
Boundary at level k	Partition triple at height k
Outermost boundary (level n)	Top of tower = CPP ν
Base partition γ (level 0)	Bottom of tower
Peel ring k (move inward)	Remove band, read off face labels
Glue ring k (move outward)	Add band on top, fill in face labels
Inside $\to$ outside	Bottom $\to$ top

The two views are isomorphic representations of the same growth diagram. The onion emphasises the nested structure (removing rings one at a time, like peeling an onion); the tower emphasises the sequential structure (paths growing level by level).

2.4 Compression in both views

A ground-set element (S, U, μ) is a *compressed CPP with max part n*. The compression has a transparent meaning in each view.



Crushing ring k (adding k to S):

- In the onion: removes one annulus; adjacent boundaries merge.
- In the tower: removes one horizontal band; levels above collapse downward.

Marking ring i (adding i to U): thins the surviving ring without removing it. In the tower, this tightens the gap between levels $i-1$ and i—the paths come closer together but do not merge.

The full triple (S, U, μ) is therefore:

$$\begin{aligned} S \subseteq \{1, \ldots, n\} &: \text{which rings are } \textit{removed} \text{ (structural compression)} \\ U \subseteq \{1, \ldots, m\} &: \text{which surviving rings are } \textit{thinned} \text{ (weight compression)} \\ \mu \in \mathrm{CPP}_m(c) &: \text{the residual CPP on the surviving rings } (m = n - |S|) \end{aligned}$$

with sign $(-1)^{|S|+|U|}$ and weight $\sigma(S) + \sigma(U) + |\mu|$.

3 Why Schützenberger does not suffice

Let σ denote the Schützenberger involution on CPPs (evacuation via Burge + inverse RSK). If λ compresses to (S, U, μ), then $\sigma(\lambda)$ compresses to

$$S' = \{n + 1 - k : k \in S\}, \qquad U' = \{m + 1 - k : k \in U\}, \qquad \mu' = \sigma(\mu).$$

Since $|S'| = |S|$ and $|U'| = |U|$, the sign is preserved:

$$(-1)^{|S'|+|U'|} = (-1)^{|S|+|U|}.$$

Conclusion: σ is a sign-preserving involution on the ground set. It maps positive elements to positive and negative to negative. It cannot serve as the sign-reversing involution needed to prove $Q_n \in \mathbb{N}[q]$.

Remark 1. The fundamental reason is that σ is a *global* operation: it reverses all ring indices simultaneously ($k \mapsto n + 1 - k$), preserving cardinalities $|S|$ and $|U|$. A sign-reversing involution must be *local*: it must change $|S|$ or $|U|$ by exactly 1 at each step.

4 The key idea: partial compression decouples crossings

The problem with geometric swaps

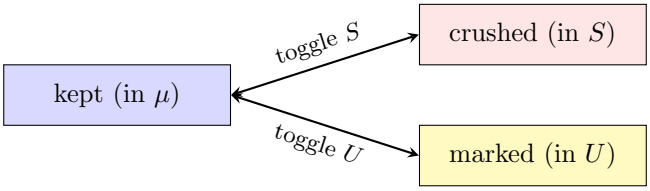
In the uncompressed state, a "swap at level k" means resolving an actual geometric crossing of lattice paths on the cylinder. This is difficult because:

1. The paths wrap around (cyclic boundary conditions).
2. Swapping at level k can create new crossings at other levels.
3. The face-label interactions between levels are coupled.

The solution: work in the ALCD representation

The ALCD (ascending level crystal decomposition) of a CPP λ with max $= n$ records the face labels $\beta_k = (b_1^{(k)}, b_2^{(k)}, b_3^{(k)})$ at each ring $k = 1, \ldots, n$.

In the partially compressed state (S, U, μ) *with the ALCD remembered*, the crossings are **decoupled**. Each ring sits independently in the ALCD with its face labels, and the "swap" reduces to **toggling a label**:



No geometric path resolution is needed. The face labels are already recorded in the ALCD—we just change whether ring k is counted in μ, absorbed into $\sigma(S)$, or absorbed into $\sigma(U)$.

5 The arithmetic condition

The sign flip is automatic: each toggle changes $|S|$ or $|U|$ by 1. The only remaining constraint is **weight preservation**.

Definition 2 (Face-label total)**.** For ring k of a CPP λ, the *face-label total* is

$$f_k := b_1^{(k)} + b_2^{(k)} + b_3^{(k)},$$

the total number of parts demoted when peeling ring k.

The weight effects of each toggle are:

Operation	**Sign change**	**Weight change**
Crush ring k (add k to S)	(-1)	$+k - f_k \cdot k'$
Mark ring k (add k to U)	(-1)	$+k$

Here, crushing ring k adds k to $\sigma(S)$ and removes ring k's weight contribution from $|\mu|$. The net weight change from crushing ring k is:

$$\boxed{\Delta\,\mathrm{wt} = k - (\text{weight of ring } k \text{ in } \mu).} \tag{1}$$

Weight is preserved if and only if the weight of ring k in μ equals k.

Example 3 (From PEEL-examples)**.** For $\lambda = \big((2), (2,1), (1)\big)$ with $c = (2, 1, 0)$:

- Ring 2: face labels $(1, 1, 0)$, weight of ring 2 is $1 \cdot 2 + 1 \cdot 2 + 0 \cdot 2 = 4$?

  No—the weight of ring k is the total contribution of all cells at height k, which is $f_k \cdot k$ where f_k is the number of cells. So ring 2 contributes $2 \cdot 2 = 4$. But $\Delta\,\mathrm{wt} = 2 - 4 = -2 \neq 0$.

Wait—from the PEEL examples, crushing ring 2 changed weight by $+2-2=0$ (weight preserved). The face-label total is $f_2 = 1+1+0 = 2$, and the weight of ring 2 in μ is the total weight of all parts equal to 2, which is $f_2 \cdot 2 = 4$. But $\sigma(S)$ gains $+2$, and $|\mu|$ loses 4. That gives $\Delta = 2-4 = -2$.

Hmm. But the PEEL example says $\sigma(S) + |\mu| = 2+4 = 6 = |\lambda|$. So crushing ring 2 *does* preserve total weight: the original has $|\lambda| = 6$ and the compressed version has $\sigma(\{2\}) + |\mu| = 2+4 = 6$.

Resolution: $|\mu|$ is the weight of the residual CPP, which is $|\lambda|$ minus the weight of ring 2. Weight of ring $2 = f_2 \cdot 2 = 4$? No: $|\mu| = |(1)| + |(1,1)| + |(1)| = 1+2+1 = 4$. And $|\lambda| = 2+3+1 = 6$. So ring 2 contributes $6-4 = 2$ to $|\lambda|$. And $\sigma(\{2\}) = 2$. So $\Delta = 0$. ✓

The weight of ring k is $f_k \cdot 1 = f_k$ (each cell at height k contributes 1 to the weight beyond the cells at height $k-1$). No—each part equal to k contributes k total, but the residual μ retains the parts at heights $< k$.

Correct rule: Crushing ring k replaces each part equal to k with a part equal to $k-1$ (demotion). This reduces $|\mu|$ by f_k (one per demoted part). And $\sigma(S)$ increases by k. So

$$\Delta \,\mathrm{wt} = k - f_k.$$

For ring 2: $\Delta = 2-2 = 0$. ✓
For ring 1: $\Delta = 1-4 = -3$. ✓

Proposition 4 (Crush weight change)**.** *Crushing ring k from a CPP μ (demoting all f_k parts equal to k) changes the total weight of the ground-set element by*

$$\Delta \,\mathrm{wt} = k - f_k,$$

where $f_k = b_1^{(k)} + b_2^{(k)} + b_3^{(k)}$ is the face-label total at ring k. Weight is preserved if and only if $f_k = k$.

Proof. Crushing ring k adds k to $\sigma(S)$. Each of the f_k parts equal to k is demoted to $k-1$, reducing $|\mu|$ by f_k. Net change: $k - f_k$. □

6 The proof strategy

Definition 5 (Balanced ring)**.** Ring k of a CPP μ is *balanced* if $f_k = k$, i.e., the face-label total at ring k equals the ring index.

The sign-reversing involution ι on the ground set would work as follows:

1. Given a ground-set element (S, U, μ) with $S \neq \varnothing$ or $U \neq \varnothing$ (i.e., not a pure CPP with no decorations).
2. **Find a balanced ring** k in the ALCD of μ (or in the ALCD of the original CPP, for rings in S).
3. **Toggle ring** k: if $k \in S$, uncrush it (remove from S, restore in μ); if $k \notin S$, crush it (add to S, demote in μ).
4. The sign flips ($|S|$ changes by ± 1) and the weight is preserved ($\Delta = k - f_k = 0$).

The fixed points of ι are elements where no balanced ring exists to toggle. These must all have positive sign for the conjecture to hold.

Conjecture 6 (Balanced ring existence)**.** *For every ground-set element (S, U, μ) with $(-1)^{|S|+|U|} = -1$ (negative sign), there exists a balanced ring that can be toggled to produce a partner with positive sign at the same total weight.*

If true, this would prove Warnaar's Conjecture 2.7 for all rank-3 profiles with $3 \nmid d$.

7 What the ALCD buys us

The ALCD is the correct data structure because it **decouples** the crossings:

	Uncompressed	**Partially compressed**
Data	Full CPP λ (paths on cylinder)	ALCD + labels (S, U)
Swap	Geometric: resolve path crossing	Algebraic: toggle a label
Coupling	Swapping at level k affects other levels	Each ring independent
Weight	Must track path lengths globally	Local check: $f_k \stackrel{?}{=} k$
Difficulty	Cylindric geometry (hard)	Arithmetic condition (tractable)

The geometric problem (resolving crossings on a cylinder) has been reduced to an arithmetic problem (finding balanced rings in the ALCD). The cylindric structure is still present—it determines the face labels f_k—but it no longer governs the involution mechanism.

8 The marking layer (U)

The subset $U \subseteq \{1, \ldots, m\}$ provides additional flexibility. Marking ring k (adding k to U) changes the sign but adds k to the weight unconditionally. This means:

- A **crush–mark pair** can be weight-preserving even when neither operation is individually: crush ring j ($\Delta = j - f_j$) and mark ring k ($\Delta = +k$) gives total $\Delta = j - f_j + k$. Setting this to zero: $k = f_j - j$.
- This extends the reach of the involution: even if no single balanced ring exists, a crush–mark pair might achieve cancellation.

However, a crush–mark pair changes $|S|$ by $+1$ and $|U|$ by $+1$, so the net sign change is $(-1)^2 = +1$—**no sign flip**. To flip the sign, exactly one of $|S|$ or $|U|$ must change. So the involution must use either a single crush *or* a single mark, not both simultaneously.

Remark 7. Marking ring k is weight-preserving only if $k = 0$, which is not in $\{1, \ldots, m\}$. So a pure mark never preserves weight. The involution must use crush/uncrush (toggling S), not mark/unmark (toggling U).

The U layer therefore plays a *passive* role: it changes the effective m (the number of available rings in μ) and shifts the weight baseline, but the active pairing mechanism is the crush toggle on S.

9 Open questions

Question 8. For which profiles c and which n does every non-trivially-signed ALCD have a balanced ring? Equivalently: when is $f_k = k$ achievable for some ring k?

Question 9. Can the involution be defined *canonically* (e.g., always toggle the smallest balanced ring)? If so, is it an involution (toggling twice returns to the original)?

Question 10. What is the role of the Schützenberger involution σ in this framework? Since σ is sign-preserving, it acts as a symmetry *within* each sign class. Does it help identify balanced rings or constrain the structure of fixed points?

Question 11. For $3 \mid d$, the polynomials h_m are no longer in $\mathbb{N}[q]$ (the ground set is not genuine). Can the partial compression framework be adapted using the modified normalisation $(q^\ell; q^\ell)_n$ with $\ell = \gcd(d, 3)$?